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SUMMARY

By day I am a software engineer, and on my own time I am an AI engineer. In my personal projects I get small, low-cost models to perform like far more expensive ones, tuning free and cheap models toward more relevant answers, an idea that grew out of my master's research on steering model responses through activation steering. At VelocitiPM I build the web app that product teams use to plan their work, sort tasks into sprints, and track progress. The project I have enjoyed most is Hermes, a personal AI agent I wired into my Google Workspace to turn a week of email, calendar, and commits into a Friday digest, plus a content creator that researches, writes, and posts to Instagram on its own.

TECHNICAL SKILLS

Languages & Runtime: Python, TypeScript, JavaScript, C++, SQL, Linux/Bash

AI / ML: PyTorch, LangGraph, Gemini API, OpenAI API, Vercel AI SDK, Activation Steering, Multi-Agent Systems, Agentic Workflow Design, RAG Architecture, Prompt Engineering, Embeddings

Stack & Infrastructure: React, Next.js, FastAPI, Node.js, Express.js, Tailwind CSS, Supabase, PostgreSQL, Redis, Vercel, AWS (EC2, S3), Docker, GitHub Actions, Playwright, Phaser 3

WORK EXPERIENCE

Software Engineer · VelocitiPM

May 2026 – Present

Full-time, Remote · Velociti is a web app that helps product teams turn ideas into plans, organize the work into sprints, and track it through to done.

- Built the team's main task board, where people drag tasks into sprints, switch between backlog, active, and completed views, and update many tasks at once. Written in React and TypeScript.
- Built the interface for Velo AI, the product's built-in AI assistant that drafts tickets and user stories for a feature, so users can run it and review the results before saving.
- Contributed to the Python AI service behind it: fed the product's design details into the context the models receive before they generate, so their output matches the intended design, and made that step work reliably across different AI providers.
- Designed slide-out side panels for linking related work, with searchable, filterable lists so users reach the right item in a couple of clicks.
- Matched each screen to the designers' Figma mockups down to the details, and wrote automated tests alongside nearly every change so existing features keep working.
- Use Claude as a daily coding partner to plan, build, and review features, which speeds up how fast I can turn a design into working, tested screens.

RESEARCH EXPERIENCE

Graduate Research: Activation Steering for Automatic Program Repair

Aug – Dec 2025

Iowa State University, Ames, IA · Advisor: Prof. Simanta Mitra · Master's Creative Component

- Improved how two open-source coding models, CodeLlama and Qwen, write code by adjusting the signals inside the model as it generates, a published method called activation steering that shapes the output without retraining. Built in PyTorch.
- Did the hands-on tuning across dozens of configurations on the university's compute cluster, dialing in the steering strength for each model and testing it against a standard coding benchmark (HumanEval) and a library of real-world software bugs (Defects4J) until the answers came out more correct and reliably ran.
- Raised the models' first-try success rate, roughly 8 percentage points on CodeLlama, and brought invalid, non-compiling output down to near zero, with the improvements holding on both model families.

Graduate Research: Secure Deep Learning & Federated Learning Fairness

Jan – May 2025

Iowa State University, Ames, IA

- Engineered ML training loops inside Intel SGX enclaves using C++ and the Darknet framework; the secure enclave added about 50% compute overhead, raising a run from 27.43s to 41.07s, with AES-GCM encrypted weight export for privacy-critical workloads.
- Formally proved fairness in a three-client federated learning protocol, showing with the NuSMV model checker and temporal logic that no client can be permanently starved and every client's updates eventually count, and that the protocol stays deadlock-free and finishes each round in bounded time.

PROJECTS

Autonomous Job Application Agent System

Python · Gemini 2.5 Pro · Apify · Playwright · Notion API · Firebase · React · TypeScript

- Five agents run the whole job search: they scout LinkedIn, Indeed, and Glassdoor, tailor a resume and cover letter per posting with Gemini 2.5 Pro, fill the forms, and pause for my confirmation before submitting.
- The submitter drives Chrome and tags page elements with Set-of-Mark, so it handles Greenhouse and Lever forms with no site-specific code. Getting the agents to cooperate around shared state was the part I learned the most from.

Bounce Streak

reddit.com/r/bounce_streak_dev

Phaser 3 · Reddit Devvit SDK · TypeScript · Express 5 · Redis · Gemini API

- A competitive 2D physics game that runs natively inside Reddit, built on Phaser 3 and the Devvit SDK. It held 60 FPS with 3,000+ players on at once.
- A daily seed hands everyone the same physics rules, so the leaderboard rewards skill over timing. I sized the control to a 38px zone to keep touch input steady on phones, where most of the play happens.

AI Social Media Generator

github.com/devendrasaim/AI-Social-Media-Generator

Python · Gemini 2.5 Flash · Imagen 3 · Perplexity API · instagrapi · Pillow

- Posts to Instagram on its own every day: it researches trends on Perplexity and YouTube, writes captions with Gemini, builds carousel images with Imagen 3, and publishes through instagrapi. No manual step anywhere.
- A self-refilling topic queue keeps it fed, a Discord webhook confirms each post, and Windows Task Scheduler starts it each morning, so the whole loop runs with no paid publishing API.

Personal AI Agent System (Hermes)

Hermes Agent · OpenRouter API · Tailscale · Gmail API · Google Calendar API · MCP · OAuth 2.0

- A local AI agent on Windows I can reach from my phone anywhere over a Tailscale VPN, connected to Gmail, Google Calendar, and GitHub through MCP and OAuth 2.0.
- I wrote a custom SKILL.md that turns a week of commits and calendar events into a Friday digest. On its first run, the agent built a working Pomodoro app by chaining 27 tools on its own.

AI English Speaking Trainer

Node.js · Express.js 5 · Gemini Interactions API · JSON Schema · Web Speech API

- A spoken-English coach with two modes, casual practice and a role-specific mock interview. An Express.js 5 backend holds the conversation server-side and scores each session on grammar, vocabulary, clarity, and relevance.
- JSON Schema locks the report format, the browser's Web Speech API handles voice with no paid service, and a three-model fallback keeps it answering when one model is down.

Portfolio Website with AI Chatbot

devendrasaim.vercel.app

Next.js · TypeScript · Vercel Serverless Functions · Gemini API · Resend · Tailwind CSS

- My live site on Vercel. The AI chatbot runs through a serverless route so Gemini keys stay off the browser, and a Resend-backed form handles contact.
- Every push to main deploys on its own and each PR gets its own preview URL first. A three-model Gemini fallback keeps the chatbot responsive when a tier is busy.

EDUCATION

Master of Science, Computer Science · Iowa State University, Ames, IA

Aug 2023 – Dec 2025

GPA: 3.5 / 4.0 · Creative Component: Automatic Program Repair Through Activation Steering (Advisor: Prof. Simanta Mitra)

Bachelor of Technology, Computer Science & Engineering · SRM University

Aug 2019 – May 2023

Amaravati, India · GPA: 7.8 / 10

CERTIFICATIONS

Prompt Design in Vertex AI · Google Cloud · Feb 2026

Cloud Infrastructure Foundations Associate · Oracle · 2021